

Ion traps

12

This chapter describes the principal methods of building ion traps and a few of their many applications in physics. The examples illustrate the extremely high-resolution spectroscopy possible with microwave and laser radiation. The theme of precision measurement in an environment with very few perturbations continues in the next chapter on quantum computing—an application that has stimulated a new wave of research on trapped ions.

12.1 The force on ions in an electric field

Charged particles in electromagnetic fields experience much larger forces than neutral atoms. An ion with a single charge $e = 1.6 \times 10^{-19} \text{ C}$ in an electric field of 10^5 V m^{-1} experiences a force

$$F_{\text{ion}} = eE \approx 10^{-14} \text{ N}. \quad (12.1)$$

This electric field corresponds to 500 V between electrodes which are 5 mm apart.¹ In comparison, a neutral atom with a magnetic moment of one Bohr magneton in a magnetic field gradient² of $\text{dB}/\text{dz} = 10 \text{ T m}^{-1}$ experiences a force of magnitude

$$F_{\text{neutral}} = \mu_{\text{B}} \left| \frac{\text{dB}}{\text{dz}} \right| \simeq 10^{-22} \text{ N}. \quad (12.2)$$

Ions feel a force 10^8 times greater than magnetically-trapped neutrals. We also see this large difference in a comparison of trap depths. In a trap operating with a voltage of $V_0 = 500 \text{ V}$, singly-charged ions have a maximum ‘binding’ energy of order 500 eV.³ This trap depth corresponds to the kinetic energy at a temperature of $6 \times 10^6 \text{ K}$. This is more than enough to trap ions, even if the ions do receive a large recoil kick during the ionization process—to load an ion trap experimenters send a weak (neutral) atomic beam through the trapping region where an electron beam ionizes a few of the atoms by knocking an electron off. These ions created by electron bombardment have much greater kinetic energy than the thermal energy of atoms at room temperature (equivalent to only 1/40 eV). It would be unwise to try to be more precise about the typical energy of an ion since it depends on the voltage used. In contrast, a magnetic trap for neutrals has a maximum depth of only 0.07 K. This was estimated in Section 10.1 by taking the magnetic energy $\mu_{\text{B}} B$ for $B = 0.1 \text{ T}$, e.g. the force in eqn 12.2 over a distance of

12.1 The force on ions in an electric field	259
12.2 Earnshaw’s theorem	260
12.3 The Paul trap	261
12.4 Buffer gas cooling	266
12.5 Laser cooling of trapped ions	267
12.6 Quantum jumps	269
12.7 The Penning trap and the Paul trap	271
12.8 Electron beam ion trap (EBIT)	275
12.9 Resolved sideband cooling	277
12.10 Summary of ion traps	279
Further reading	279
Exercises	280

¹This assumes electrodes in the form of a parallel-plate capacitor. Although ion traps have a different geometry, this still gives a reasonable estimate and shows that the electrostatic force gives strong trapping for a voltage readily available in the laboratory.

²This value is typical of magnetic traps with coils wound with copper wire. Superconducting magnets give higher gradients.

³We consider ions with a single positive charge $+e$ such as Mg^+ , Ca^+ and Hg^+ , since few experiments use species that acquire additional electrons to give negative ions. Section 12.8 deals with highly-charged ions.

10 mm. These estimates show that neutral atoms must be cooled before trapping but ion trapping requires only moderate electric fields to capture the charged particles directly. It is not straightforward, however, to find a suitable electric field configuration and, as in many advances within atomic physics, the success of ion trapping relies on some subtle ideas rather than a brute-force approach.

12.2 Earnshaw's theorem

Earnshaw proved that: *A charge acted on by electrostatic forces cannot rest in stable equilibrium in an electric field.*⁴

⁴The theorem dates back to the nineteenth century and James Clerk Maxwell discussed it in his famous treatise on electromagnetism.

⁵The derivation of this equation from the Maxwell equation $\text{div } \mathbf{D} = \rho_{\text{free}}$ assumes $\rho_{\text{free}} = 0$ and a linear isotropic homogeneous medium in which $\mathbf{D} = \epsilon_r \epsilon_0 \mathbf{E}$ with ϵ_r constant. Ions are usually trapped in a vacuum, where $\epsilon_r = 1$.

Thus it is not possible to confine an ion using a purely electrostatic field. Physicists have invented ingenious ways around this theorem but, before describing the principles of these ion traps, we need to think about the underlying physics. The theorem follows from the fact that an electric field has no divergence in a region with no free charge density, $\text{div } \mathbf{E} = 0$.⁵ Zero divergence means that all the field lines going into a volume element must come out—there are no sources or sinks of field within the volume. Equivalently, Gauss' theorem tells us that the integral of the normal component of $\mathbf{E}$ over the bounding surface equals the volume integral of $\text{div } \mathbf{E}$, which is zero:

$$\oiint \mathbf{E} \cdot d\mathbf{S} = \iiint \text{div } \mathbf{E} \, d^3\mathbf{r} = 0. \quad (12.3)$$

Hence $\mathbf{E} \cdot d\mathbf{S}$ cannot have the same sign over all of the surface. Where $\mathbf{E} \cdot d\mathbf{S} < 0$ the electric field points inwards and a positive ion feels a force that pushes it back into the volume; but $\mathbf{E} \cdot d\mathbf{S} > 0$ somewhere else on the surface and the ion escapes along that direction. A specific example of this is shown in Fig. 12.1 for the field produced by two equal positive charges with a fixed separation along the z -axis. Midway between the charges, at the point labelled P, the electric fields from the two charges cancel and the ion experiences no force, but this does not give stable equilibrium. The argument above holds true, so the electric field lines around the point P cannot all be directed inwards. When slightly displaced from P, a positive ion accelerates perpendicular to the axis, whereas a negative ion would be attracted towards one of the fixed charges. This behaviour can also be explained by the fact that the point P is a saddle point of the electrostatic potential ϕ . The electrostatic potential energy $e\phi$ of the ion has the same form as the gravitational potential energy of a ball placed on the saddle-shaped surface shown in Fig. 12.2—clearly the ball tends to roll off down the sides. In this alternative way of looking at Earnshaw's theorem in terms of electrostatic potential rather than the fields, stable trapping does not occur because the potential never has a minimum, or maximum, in free space.⁶

⁶It takes just a few lines of algebra to prove this from Laplace's equation.

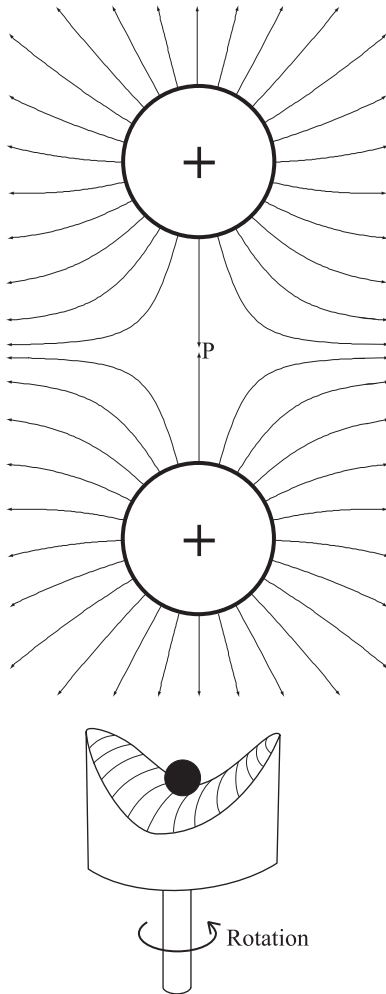


Fig. 12.1 The electric field lines between two equal positive charges. Midway between the charges at the point P the electric fields from the two charges cancel. At this position the ion experiences no force but it is not in stable equilibrium. At all other positions the resultant electric field accelerates the ion.

Fig. 12.2 A ball on a saddle-shaped surface has a gravitational potential energy that resembles the electrostatic potential energy of an ion in a Paul trap. Rotation of the surface about a vertical axis, at a suitable speed, prevents the ball rolling off the sides of the saddle and gives stable confinement.

12.3 The Paul trap

The analogy with a ball moving on the saddle-shaped surface shown in Fig. 12.2 provides a good way of understanding the method for confining ions invented by Wolfgang Paul. The gravitational potential energy of the ball on the surface has the same form as the potential energy of an ion close to a saddle point of the electrostatic potential. We assume here a symmetric saddle whose curvature has the same magnitude, but opposite sign, along the principal axes:

$$z = \frac{\kappa}{2} \left[(x')^2 - (y')^2 \right], \quad (12.4)$$

where $x' = r \cos \Omega t$ and $y' = r \sin \Omega t$ are coordinates in a frame rotating with respect to the laboratory frame of reference. The time average of this potential is zero. Rotation of the saddle shape around the vertical axis turns the unstable situation into stable mechanical equilibrium, and

makes an impressive lecture demonstration. Such dynamic stabilisation cannot honestly be described as ‘well known’ so it is explained carefully here by an approximate mathematical treatment of ions in an a.c. field.

12.3.1 Equilibrium of a ball on a rotating saddle

In the mechanical analogue, the rotation of the saddle at a suitable speed causes the ball to undergo a wobbling motion; the ball rides up and down over the low-friction surface of the rotating saddle shape and the ball’s mean position only changes by a small amount during each rotation.⁷ The amplitude of this wobbling increases as the ball moves further from the centre of the saddle. For this oscillatory motion the time-averaged potential energy is not zero and the total energy (potential plus kinetic) increases as the object moves away from the centre. Therefore the mean position of the object (averaged over many cycles of the rotation) moves as if it is in an effective potential that keeps the ball near the centre of the saddle. We will find that an ion jiggling about in an a.c. field has a similar behaviour: a fast oscillation at a frequency close to that of the applied field and a slower change of its mean position.

12.3.2 The effective potential in an a.c. field

To explain the operation of the Paul trap, we first look at how an ion behaves in a.c. electric field $\mathbf{E} = \mathbf{E}_0 \cos(\Omega t)$. An ion of charge e and mass M feels a force $\mathbf{F} = e\mathbf{E}_0 \cos(\Omega t)$, and so Newton’s second law gives

$$M\ddot{\mathbf{r}} = e\mathbf{E}_0 \cos(\Omega t). \quad (12.5)$$

Two successive integrations give the velocity and displacement as

$$\begin{aligned} \dot{\mathbf{r}} &= \frac{e\mathbf{E}_0}{M\Omega} \sin(\Omega t), \\ \mathbf{r} &= \mathbf{r}_0 - \frac{e\mathbf{E}_0}{M\Omega^2} \cos(\Omega t). \end{aligned} \quad (12.6)$$

It has been assumed that the initial velocity is zero and $\mathbf{r}_0$ is a constant of integration. The field causes the ion to oscillate at angular frequency Ω with an amplitude proportional to the electric field. From this steady-state solution we see that the forced oscillation does not heat the ions.⁸ (These simple steps form the first part of the well-known derivation of the *plasma frequency* for a cloud of electrons in an a.c. field, given in most electromagnetism texts.) The following section describes an example of this behaviour in which the amplitude of the electric field changes with position $\mathbf{E}_0(\mathbf{r})$.

12.3.3 The linear Paul trap

In a linear Paul trap the ion moves in the field produced by the electrodes shown in Fig. 12.3. The four rods lie parallel to the z -axis and at the corners of a square in the xy -plane. Each electrode is connected to the

⁷Although we shall not analyse the mechanical system in detail, it is important to note that the wobbling motion is not entirely up and down but has radial and tangential components. Similarly, an object that floats on the surface of water waves does not just bob up and down but also oscillates back and forth along the direction of propagation of the waves, so its overall motion in space is elliptical. The discussion only applies for mechanical systems where friction has a negligible effect so that the ball slides smoothly over the surface.

⁸The a.c. field does not change the ion’s average total energy because the work done on the ion given by $\mathbf{F} \cdot \dot{\mathbf{r}} \propto \cos(\Omega t) \sin(\Omega t)$ averages to zero over one cycle—the force and velocity have a phase difference of $\pi/2$.

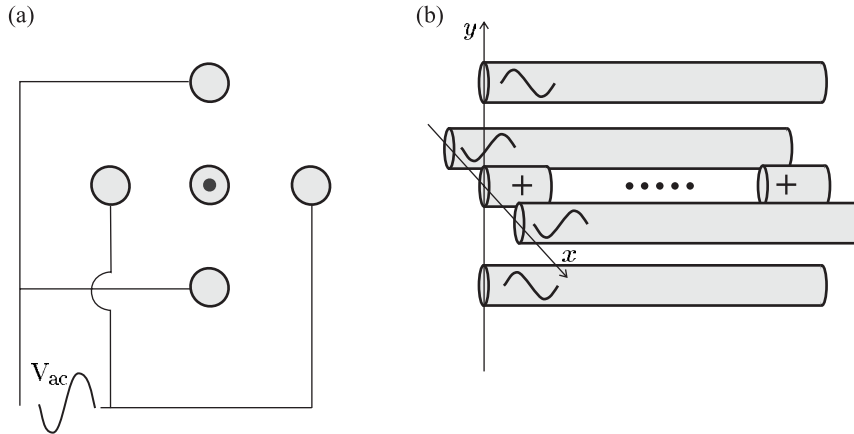


Fig. 12.3 A linear Paul trap used to store a string of ions. (a) A view looking along the four rods with the end-cap electrode and ions in the centre. Each of the rods is connected to the one diagonally opposite so that a voltage between the pairs gives a quadrupole field. (b) A side view of the rod and end-cap electrodes which have a.c. and positive d.c. voltages, respectively. A string of trapped ions is indicated.

one diagonally opposite and the a.c. voltage $V = V_0 \cos(\Omega t)$ is applied between the two pairs. Despite the fact that the voltages vary with time, we first find the potential by the usual method for electrostatic problems. The electrostatic potential ϕ satisfies Laplace's equation $\nabla^2 \phi = 0$ (because $\text{div } \mathbf{E} = 0$ and $\mathbf{E} = -\nabla \phi$). A suitable solution for the potential close to the z -axis, that matches the symmetry of the voltages on the electrodes, has the form of a quadrupole potential

$$\phi = a_0 + a_2(x^2 - y^2). \quad (12.7)$$

The coefficients a_0 and a_2 are determined from the boundary conditions. There are no terms linear in x or y because of the symmetry under reflection in $x = 0$ and $y = 0$. The terms in x^2 and y^2 have opposite signs, and the variation with z is negligible for rods much longer than their separation $2r_0$. The potential must match the boundary conditions

$$\begin{aligned} \phi &= \phi_0 + \frac{V_0}{2} \cos(\Omega t) & \text{at} & \quad x = \pm r_0, y = 0, \\ \phi &= \phi_0 - \frac{V_0}{2} \cos(\Omega t) & \text{at} & \quad x = 0, y = \pm r_0. \end{aligned} \quad (12.8)$$

These conditions are satisfied by the potential⁹

$$\phi = \phi_0 + \frac{V_0}{2r_0^2} \cos(\Omega t) (x^2 - y^2). \quad (12.9)$$

To solve Laplace's equation we have simply made a reasonable guess, taking into account the symmetry. This is perfectly justified since the uniqueness theorem says that a solution that fits the boundary conditions is the only valid solution (see electromagnetism texts). The usual

⁹This ignores the finite size of the electrodes and that the inner surfaces of the electrodes would need to be hyperbolic, e.g. a surface given by $x^2 - y^2 = \text{const.}$, to match the equipotentials. However, by symmetry this potential has the correct form for $r \ll r_0$, no matter what happens near to the electrodes.

¹⁰This method would not be appropriate for shorter wavelengths, e.g. microwave radiation with frequencies of GHz.

¹¹The two-dimensional quadrupole field between the four rods looks *superficially* like the quadrupole field with cylindrical symmetry in Fig. 12.1 and the analogy with a rotating saddle applies to both. A comparison of the potentials for the two cases in eqn 12.9 and eqn 12.23 shows that they are different. Note also that the electrostatic potential oscillates ‘up and down’ rather than rotating as in the mechanical analogy.

¹²The Mathieu equation arises in a variety of other physical problems, e.g. the inverted pendulum. A pendulum is normally considered as hanging down from its pivot point and undergoing simple harmonic motion with a small amplitude. In an inverted pendulum a rod, that is pivoted at one end, initially points vertically upwards; any slight displacement from this unstable position causes the rod to fall and swing about the stable equilibrium position (pointing straight down), but if the pivot point oscillates rapidly up and down then the rod remains upright whilst executing a complicated motion—the rod can make quite large-angle excursions from the vertical direction without falling over. The mathematical textbook by Acheson (1997) gives further details of the complexities of this fascinating system and numerical simulations can be seen on the web site associated with that book.

¹³This corresponds to the motion of an ion in a trap that has no d.c. voltage. In practice, ion traps may have some d.c. voltage because of stray electric fields but this can be cancelled by applying a suitable d.c. voltage to the electrodes (or additional electrodes near the four rods). The solution of the Mathieu equation with $a_x \neq 0$ is discussed in the book on ion traps by Ghosh (1995).

¹⁴More detailed mathematical treatments of the Mathieu equation can be found in Morse and Feshbach (1953) and Mathews and Walker (1964).

method for solving an electrostatic problem applies even though the voltage on the electrodes changes because at the radio-frequencies (used in ion traps) the radiation has a wavelength much greater than the dimensions of the electrodes, e.g. a wavelength of 30 m for $\Omega = 2\pi \times 10$ MHz.¹⁰ The potential energy $e\phi$ of an ion has a saddle point in the middle of these electrodes that looks like the saddle shape shown in Fig. 12.2—a potential ‘hill’ in the x -direction and a ‘valley’ in the y -direction, or the other way around.¹¹ From the gradient of potential we find the electric field

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_0(\mathbf{r}) \cos(\Omega t) \\ &= -\frac{V_0}{r_0^2} \cos(\Omega t) (x\hat{\mathbf{e}}_x - y\hat{\mathbf{e}}_y). \end{aligned} \quad (12.10)$$

The equation of motion in the x -direction is

$$M \frac{d^2 x}{dt^2} = -\frac{eV_0}{r_0^2} \cos(\Omega t) x. \quad (12.11)$$

A change of variable to $\tau = \Omega t/2$ leads to

$$\frac{d^2 x}{d\tau^2} = -\frac{4eV_0}{\Omega^2 M r_0^2} \cos(2\tau) x. \quad (12.12)$$

This is a simplified form of the Mathieu equation:¹²

$$\frac{d^2 x}{d\tau^2} + (a_x - 2q_x \cos 2\tau) x = 0 \quad (12.13)$$

with $a_x = 0$.¹³ It is conventional to define the parameter in front of the oscillating term as $2q_x$ (in anticipation of this e has been used for the ion’s charge), where

$$q_x = \frac{2eV_0}{\Omega^2 M r_0^2}. \quad (12.14)$$

We look for a solution of the form

$$x = x_0 \cos A\tau \{1 + B \cos 2\tau\}. \quad (12.15)$$

The arbitrary constant A gives the angular frequency of the overall motion and B is the amplitude of the fast oscillation at close to the driving frequency. The justification for choosing this form is that we expect an oscillating driving term to produce a periodic solution and substitution of a function containing $\cos A\tau$ into the equation leads to terms with $\cos A\tau \cos 2\tau$.¹⁴ Substitution into the equation (with $a_x = 0$) gives

$$\begin{aligned} x_0 [-4B \cos A\tau \cos 2\tau + 4AB \sin A\tau \sin 2\tau - A^2 \cos A\tau \{1 + B \cos 2\tau\}] \\ = 2q_x x_0 \cos 2\tau \cos A\tau \{1 + B \cos 2\tau\}. \end{aligned} \quad (12.16)$$

We shall assume that $A \ll 1$, so that the function $\cos A\tau$ corresponds to a much slower oscillation than $\cos 2\tau$, and also that the amplitude $B \ll 1$ (both of these assumptions are discussed below). Thus the terms

proportional to $\cos A\tau \cos 2\tau$ dominate on each side, and equating their coefficients gives $-4B = 2q_x$ or

$$B = -\frac{q_x}{2} = -\frac{eV_0}{M\Omega^2 r_0^2}. \quad (12.17)$$

This amplitude of the fast oscillation is consistent with the result for a uniform electric field in eqn 12.6.¹⁵ This fast oscillation is called the **micromotion**. To determine the angular frequency A we consider how the mean displacement changes on a time-scale longer than the micromotion: the time-average of $\cos^2 2\tau = 1/2$ so eqn 12.16 yields $-A^2 \cos A\tau = q_x B \cos A\tau$;¹⁶ hence $A = q_x/\sqrt{2}$ and an approximate solution is

$$x = x_0 \cos\left(\frac{q_x \tau}{\sqrt{2}} + \theta_0\right) \left\{1 + \frac{q_x}{2} \cos 2\tau\right\}, \quad (12.18)$$

where q_x is defined in eqn 12.14.¹⁷ We assumed that $q_x \ll 1$ but it turns out that this approximation works better than 1% for $q_x \leq 0.4$ (Wuerker *et al.* 1959). Since $\tau = \Omega t/2$ the mean displacement undergoes simple harmonic motion at an angular frequency given by

$$\omega_x = \frac{q_x \Omega}{2\sqrt{2}} = \frac{eV_0}{\sqrt{2}\Omega M r_0^2}. \quad (12.19)$$

A more detailed treatment shows that ions remain trapped for

$$q_x \leq 0.9 \quad (12.20)$$

or $\omega_x \leq 0.3\Omega$. For a radio-frequency field oscillating at $\Omega = 2\pi \times 10$ MHz the ion must have a radial oscillation frequency $\omega_x \leq 2\pi \times 3$ MHz. If we choose $\omega_x = 2\pi \times 1$ MHz (a convenient round number) then eqn 12.19 gives the numerical values $V_0 = 500$ V and $r_0 = 1.9$ mm for trapping Mg^+ ions.¹⁸ To get this high trapping frequency the ion trap has electrodes closer together than we assumed in the introduction. By symmetry, the same considerations apply for motion in the y -direction, and so we define a radial frequency $\omega_r \equiv \omega_x = \omega_y$. The Paul trap has a sharp transition from stable trapping to no trapping in the radial direction when q_r equals the maximum value of q_x in eqn 12.20. Paul used this feature to determine the charge-to-mass ratio e/M of the ions and hence perform mass spectroscopy—generally the charge state is known (e.g. it is e or $2e$, etc.) and hence M is determined.

So far we have only described confinement in the xy -plane. There are several ways to extend trapping to all three directions. For example, Fig. 12.3(b) shows a trap with two additional electrodes at $z = \pm z_0$ that repel the ions. For positive ions both of these end-cap electrodes have the same positive voltage to give a field similar to that shown in Fig. 12.1, with a minimum in the electrostatic potential along the z -axis—in the radial direction the static potential has a negligible effect compared to the a.c. trapping. When the linear Paul trap has axial confinement weaker than that in the radial direction, i.e. $\omega_z \ll \omega_x = \omega_y$, the ions tend to lie in a string along the z -axis with only a small micromotion

¹⁵Equation 12.10 shows that the component of the electric field in this direction is $\mathbf{E}_0(\mathbf{r}) \cdot \hat{\mathbf{e}}_x = -V_0 x/r_0^2$.

¹⁶The term $4AB \sin A\tau \cos 2\tau$ time-averages to zero.

¹⁷An arbitrary initial phase θ_0 has been included to make the expression more general but this does not affect the argument above.

¹⁸In comparison, neutral atoms in magnetic traps oscillate at frequencies in the range 10–1000 Hz.

in the radial direction; there is no axial micromotion because it is a d.c. electric field along the z -direction. We will see that after cooling the ions sit very close to the trap axis where the a.c. field causes very small perturbations. In the later discussion of laser cooling the trapped ions are considered simply as harmonic oscillators (neglecting micromotion).

12.4 Buffer gas cooling

Trapping of ions requires a vacuum, but the presence of a small background of helium gas at a pressure ($\sim 10^{-4}$ mbar) gives very effective cooling of hot ions. The ions dissipate their kinetic energy through collisions with the buffer gas atoms and this quickly brings the ions into thermal equilibrium with the gas at room temperature. For ions that start off above room temperature the buffer gas cooling actually *reduces* the perturbations on the ions. Any slight broadening and shift of the ions' energy levels by the collisions is outweighed by the reduction in the ions' micromotion—the ions stay closer to the trap centre where they see smaller a.c. fields.

Buffer gas cooling can be compared to having a vacuum flask that has partially 'lost its vacuum'—hot coffee in the flask cools down because the low pressure of gas between the walls provides much less thermal insulation than a good vacuum. It follows from this argument that ions can only be cooled far below room temperature in a good vacuum, e.g. a pressure of 10^{-11} mbar in the laser cooling experiments described later (otherwise collisions with the hot background gas atoms heat the ions). This case corresponds to having a good vacuum flask; dewars in the laboratory work on the same principle to keep things such as liquid helium at temperatures much colder than the surroundings.

Buffer gas cooling finds widespread application in instruments that need to operate reliably over long periods, such as the mercury ion clock developed at NASA's Jet Propulsion Laboratory in Pasadena, California. The linear Paul trap contains a cloud of mercury ions and microwaves drive the transition at 40 GHz between the two hyperfine levels in the ground state of the ions. By reference to the resonance frequency of the ions, the electronic servo-control system maintains the frequency of the microwave source stable to 1 part in 10^{14} over long periods. This ion trap provides a very good frequency reference and has been used for navigation in deep space, where the accurate timing of signals transmitted to and from the probe determine its distance from a transmitter/receiver on Earth.¹⁹ Paul traps with buffer gas cooling are also used to give long ion storage times in some commercial mass spectrometers.

There is a method analogous to buffer gas cooling that works at much lower temperatures. In **sympathetic cooling** a trap confines two species of ions at the same time, e.g. Be^+ and Mg^+ . Laser cooling one species, as described in the next section, e.g. the Be^+ ions, produces a cloud of cold ions that acts as the 'buffer gas' to cool the Mg^+ ions through collisions. The ions interact through their strong long-range Coulomb

¹⁹See Berkeland *et al.* (1998) for a description of a state-of-the-art laser-cooled mercury ion frequency standard.

repulsion, so they never come close enough together to react—in this sense the collisions between ions are benign (whereas neutral atoms in magnetic traps undergo some inelastic collisions leading to trap loss).

12.5 Laser cooling of trapped ions

The cooling of ions uses the same scattering force as the laser cooling of neutral atoms, and historically David Wineland and Hans Dehmelt proposed the idea of laser cooling for ions before any of the work on neutral atoms. The long confinement time in ion traps makes experiments straightforward in principle; but, in practice, ions have resonance lines in the blue or ultraviolet regions, so they often require more complicated laser systems than for neutral atoms with resonance lines at longer wavelengths.²⁰ This difference arises because in ions the valence electron sees a more highly-charged core than in the isoelectronic neutral atoms, i.e. the atom with the same electronic configuration. The shorter wavelengths for ions also means that they generally have larger natural widths than neutral atoms since Γ depends on the cube of the transition frequency. This high scattering rate for resonant laser light leads to a strong radiative force on the ions and also allows the detection of single ions, as shown below.

Each trapped ion behaves as a three-dimensional simple harmonic oscillator but a single laser beam damps the motion in all directions. To achieve this, experimenters tune the laser frequency slightly below resonance (red frequency detuning, as in the optical molasses technique), so the oscillating ion absorbs more photons as it moves towards the laser beam than when it moves away. This imbalance in the scattering during the oscillations slows the ion down. This Doppler cooling works in much the same way as in optical molasses but there is no need for a counter-propagating laser beam because the velocity reverses direction in a bound system. The imbalance in scattering arises from the Doppler effect so the lowest energy is the Doppler cooling limit $k_B T = \hbar \Gamma / 2$ (see Exercise 12.1). To cool the ion's motion in all three directions the radiative force must have a component along the direction corresponding to each degree of freedom, i.e. the laser beam does not go through the trap along any of the axes of symmetry. During laser cooling the spontaneously emitted photons go in all directions and this strong fluorescence enables even single ions to be seen! Here I do not mean detectable, but actually seen with the naked eye; a Ba^+ ion with a visible transition appears as a tiny bright dot between the electrodes when you peer into the vacuum system. The resonance transition in the barium ion has a longer wavelength than most other ions at 493 nm in the green region of the visible spectrum. More generally, experiments use CCD cameras to detect the blue or ultraviolet radiation from the ions giving pictures such as Fig. 12.4. To calculate the signal from a Ca^+ ion with a transition at a wavelength of 397 nm and $\Gamma = 2\pi \times 23 \times 10^6 \text{ s}^{-1}$, we use eqn 9.3 with

²⁰The generation of continuous-wave radiation at 194 nm for laser cooling Hg^+ requires several lasers and frequency mixing by nonlinear optics, but nowadays radiation at a wavelength of 397 nm for laser cooling Ca^+ is produced by small semiconductor diode lasers.

Fig. 12.4 A string of calcium ions in a linear Paul trap. The ions have an average separation of $10\ \mu\text{m}$ and the strong fluorescence enables each ion to be detected individually. The minimum size of the image for each ion is determined by the spatial resolution of the imaging system. Courtesy of Professor Andrew Steane and co-workers, Physics department, University of Oxford.

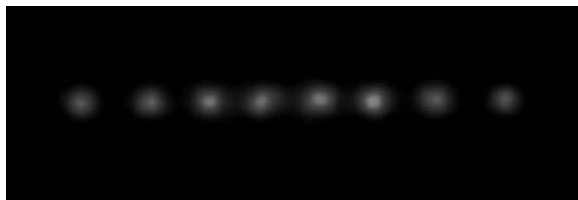
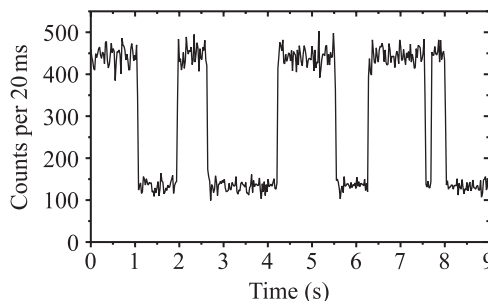


Fig. 12.5 The fluorescence signal from a single calcium ion undergoing quantum jumps. The ion gives a strong signal when it is in the ground state and it is ‘dark’ while the ion is shelved in the long-lived metastable state. Data courtesy of Professors Andrew Steane and Derek Stacey, David Lucas and co-workers, Physics department, University of Oxford.



$\delta = -\Gamma/2$ and $I = 2I_{\text{sat}}$, so that

$$\mathcal{R}_{\text{scatt}} \simeq \frac{\Gamma}{4} \simeq 4 \times 10^7 \text{ photons s}^{-1}. \quad (12.21)$$

In a typical experiment the lens that images the fluorescence onto the detector has an f -number (ratio of focal length to diameter) of about 2, so it collects 1.6% of the total number of fluorescence photons (the solid angle subtended over 4π). A reasonable detector could have an efficiency of 20%, giving an experimental signal of $S = 0.016 \times 0.2 \times \mathcal{R}_{\text{scatt}} = 10^5 \text{ photons s}^{-1}$ that can easily be measured on a photomultiplier as in Fig. 12.5. (The signal is lower than the estimate because fluorescent photons are lost by reflection at the surfaces of optical elements, e.g. windows or lenses.)

Laser cooling on a strong transition reduces the ion’s energy to the Doppler cooling limit $\hbar\Gamma/2$. In a trap with a spacing of $\hbar\omega_{\text{trap}}$ between vibrational energy levels, the ions occupy about $\sim \Gamma/\omega_{\text{trap}}$ vibrational levels; typically this corresponds to many levels. The minimum energy occurs when the ions reside in the lowest quantum level of the oscillator where they have just the zero-point energy of the quantum harmonic oscillator. To reach this fundamental limit experiments use more sophisticated techniques that drive transitions with narrow line widths, as described in Section 12.9.

12.6 Quantum jumps

In addition to the strong resonance transition of natural width Γ used for laser cooling, ions have many other transitions and we now consider excitation of a weak optical transition with a natural width Γ' , where $\Gamma' \ll \Gamma$. Figure 12.6 shows both of the transitions and the relevant energy levels. The first application described here simply uses a narrow transition to measure the temperature accurately. We shall look at high-resolution laser spectroscopy later.

A simple calculation shows that probing on a narrow transition is necessary to measure the temperature of ions at the Doppler cooling limit of the strong transition. Multiplying the velocity spread in eqn 9.29 by a factor of $2/\lambda$, as in eqn 6.38, gives the Doppler-broadening at the Doppler cooling limit as

$$\Delta f_D \simeq \frac{2v_D}{\lambda} = 2\sqrt{\frac{\hbar\Gamma}{M\lambda^2}}. \quad (12.22)$$

For the transition in the calcium ion whose parameters were given in the previous section this evaluates to $\Delta f_D = 2$ MHz, which is only 0.07 times the natural width ($\Delta f_N = 23$ MHz). Therefore the line has a width only slightly greater than the natural width and measurements of this line width cannot determine the temperature accurately. This difficulty disappears for a much narrower transition where Doppler broadening dominates the observed line width, but this scheme presents a practical problem—the low rate of scattering of photons on the weak transition makes it hard to observe the ion.

Ion trappers have developed a clever way to simultaneously obtain a good signal and a narrow line width. This experimental technique requires radiation at two frequencies so that both a strong and weak transition can be excited. When both laser frequencies illuminate the ions the fluorescence signal looks like that shown in Fig. 12.5. The fluorescence switches off and on suddenly at the times when the valence electron jumps up or down from the long-lived excited state—the average time between the switching in this *random telegraph signal* depends on the lifetime of the upper state. These are the quantum jumps between allowed energy levels postulated by Bohr in his model of the hydrogen

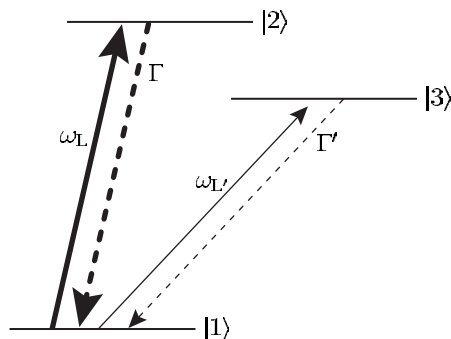


Fig. 12.6 Three energy levels of an ion. The allowed transition between levels 1 and 2 gives a strong fluorescence signal when excited by laser light. The weak transition between 1 and 3 means that level 3 has a long lifetime and $\Gamma' \ll \Gamma$ (a metastable state).

atom, and the direct observation of such jumps for a single ion provides new ways of testing quantum mechanics. Previous experiments measured the ensemble average of an observable, i.e. its average value for a collection of particles, but ion traps allow repeated measurements on a single object. In quantum language, each absorption and emission of a photon on the strong transition constitutes a measurement of the state of the ion—it is found to be in either the ground state or the long-lived excited state, and these two outcomes correspond to the two eigenvalues of the observable. The fluorescence signal in Fig. 12.5 represents a sequence of such measurements that gives a record of the state of the ion. In addition to giving an insight into the nature of quantum mechanics, the excitation of very narrow resonances has a very practical use in optical frequency standards.

The observation of a highly forbidden transition in the ytterbium ion at the National Physical Laboratory at Teddington, London furnishes an extreme example of high-resolution spectroscopy. An $^2F_{7/2}$ level in Yb^+ can only decay to the ground level $^2S_{1/2}$ by an octupole transition²¹ with a calculated natural lifetime of 10 years (Roberts *et al.* 1997). Even though excited ions are forced to decay more quickly to make experiments feasible, the rate of spontaneous emission on this transition is tiny. To detect this weak transition experiments use the scheme shown in Fig. 12.6. In addition to the weak $^2S_{1/2}$ – $^2F_{7/2}$ transition between levels 1 and 3 there is a strong $^2S_{1/2}$ – $^2P_{1/2}$ transition between levels 1 and 2. Laser radiation at frequency $\omega_{L'}$ drives the weak transition for a time t_w ; then laser light resonant with a strong transition ($\omega_L \simeq \omega_{12}$) turns on for a period t_{det} to determine the state of the ion. If the ion was excited to the long-lived upper level during t_w then it does not fluoresce. If it remains in the ground state, however, photons arrive at the detector at a rate $\mathcal{R}_{\text{obs}} \simeq 10^5 \text{ s}^{-1}$, as estimated below eqn 12.21 (for a strong transition in a calcium ion). In a period $t_{\text{det}} = 2 \times 10^{-2} \text{ s}$ there are $\mathcal{R}_{\text{obs}} t_{\text{det}} = 2000$ photons detected. Repetition of these two stages of the measurement procedure as ω_L scans over the frequency range around the narrow resonance at ω_{13} produces a plot of the probability of exciting the narrow transition (in time t_w) versus laser frequency. The line width of the observed resonance depends on the measurement time, since other broadening mechanisms are negligible. Fourier transform theory shows that the frequency width of a pulse is inversely proportional to its duration, $\Delta\omega_{\text{obs}} \sim 1/t_w$ —the same relation used in transit-time broadening (eqn 7.50). This corresponds to counting the number of cycles of the laser radiation, $f_{L'} t_w$ to within $\pm \frac{1}{2}$ of a cycle ($f_{L'} = \omega_{L'}/2\pi$). Frequency standards use this detection scheme with alternate periods of probing and measurement because the weak transition must not be perturbed by having a strong interaction at the same time.²²

²¹This is an E3 transition in notation, where E1 and E2 signify electric dipole and quadrupole transitions, respectively.

²²The logic of this method is reminiscent of a case solved by the famous detective Sherlock Holmes. From the observation that the dog did not bark in the night he deduced that the murderer was not a stranger. It is deduced that the ion has been excited when it is not observed.

12.7 The Penning trap and the Paul trap

For experiments with several ions the linear Paul trap has the advantage that the ions lie like a string of beads along the trap axis, with little micromotion. However, most of the pioneering experiments on ion traps used the electrode configuration shown in Fig. 12.7. The cylindrical symmetry of these electrodes imposes boundary conditions on the electrostatic potential, as in Section 12.3.3. Solutions of Laplace's equation that satisfy these conditions have the form

$$\phi = \phi_0 + a_2 \left(z^2 - \frac{x^2 + y^2}{2} \right). \quad (12.23)$$

The surfaces of constant potential have a hyperbolic cross-section in any radial plane, e.g. $y = 0$; in many experiments the electrodes match the shape of these equipotentials so that the potential corresponds to eqn 12.23 right out to the electrode surfaces, but any cylindrically-

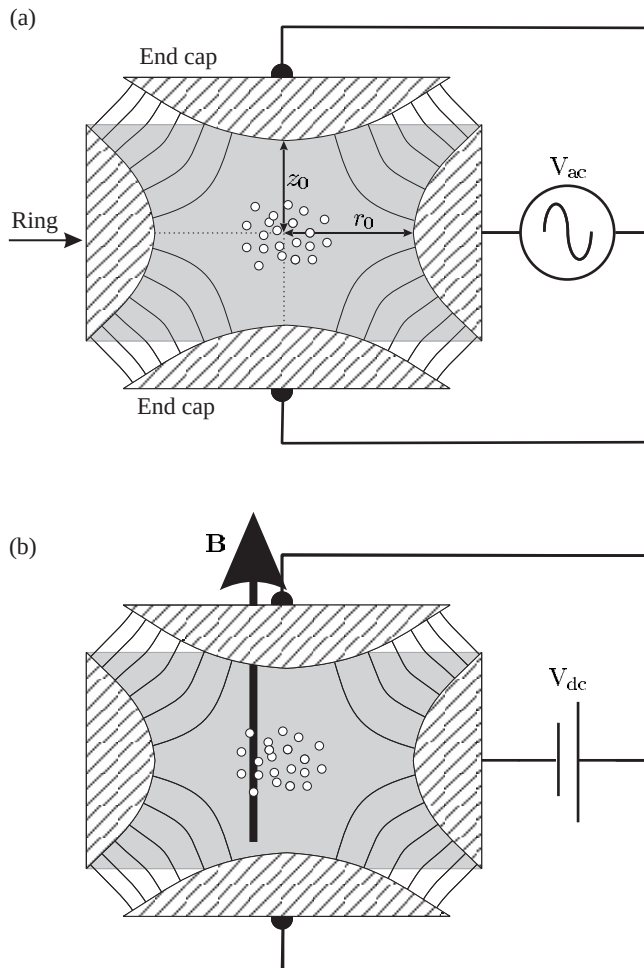


Fig. 12.7 The electrode configuration of (a) the Paul trap and (b) the Penning trap, shown in cross-section. The lines between the end caps and ring electrode indicate the electric field lines; the Paul trap has an oscillating electric field but the Penning trap has static electric and magnetic fields. The electrodes shown have a hyperbolic shape (hyperbolae rotated about the z -axis), but for a small cloud of ions confined near to the centre any reasonable shape with cylindrical symmetry will do. Small ion traps with dimensions ~ 1 mm generally have simple electrodes with cylindrical or spherical surfaces (cf. Fig. 12.3). Courtesy of Michael Nasse.

²³The higher-order terms x^4 , x^2y^2 , etc. have negligible influence near the origin.

symmetric electrodes work well for laser cooling experiments since the cold ions remain close to the trap centre, where symmetry constrains the potential to have the form of eqn 12.23.²³ An a.c. voltage between the ring electrode and the two end caps gives a Paul trap that works on the same principle as the two-dimensional trapping described in Section 12.3.3. For cylindrical symmetry the trap has an electric field gradient along z whose magnitude is twice as large as the gradient along x or y . This difference from the linear quadrupole field means that the region of stable Paul trapping occurs at slightly different voltages (values of the parameter q , defined in eqn 12.14), as described in the book on ion traps by Ghosh (1995).

The Paul trap principle can be used to confine charged particles in air, e.g. dust, or charged droplets of glycerin of diameter 10–100 μm . In this demonstration apparatus, the strong damping of the motion by the air at atmospheric pressure helps to confine charged particles over a large range of a.c. voltages (see Nasse and Foot (2001), which contains references to earlier work). In comparison, the well-known Millikan oil drop experiment just levitates particles by balancing gravity with an electrostatic force.

12.7.1 The Penning trap

The Penning trap has the same electrode shape as the Paul trap (as shown in Fig. 12.7) but uses static fields. The Paul trap is generally assumed to have cylindrical symmetry unless specifically stated as being a linear Paul trap. In a Penning trap for positive ions, both end caps have the same positive voltage (with respect to the ring electrode) to repel the ions and prevent them escaping along the axis (cf. the trap in Fig. 12.3). With only a d.c. electric field the ions fly off in the radial direction, as expected from Earnshaw's theorem, but a strong magnetic field along the z -axis confines the ions. The effect of this axial magnetic field can be understood by considering how a charged particle moves in crossed electric and magnetic fields. In an electric field $\mathbf{E} = E\hat{\mathbf{e}}_x$ the force $\mathbf{F} = e\mathbf{E}$ accelerates the positive ion along the x -axis. In a region where there is a uniform magnetic field $\mathbf{B} = B\hat{\mathbf{e}}_z$ the Lorentz force $\mathbf{F} = e\mathbf{v} \times \mathbf{B}$ causes the ion to execute circular motion at the cyclotron frequency²⁴

$$\omega_c = \frac{eB}{M}. \quad (12.24)$$

In a region of crossed electric and magnetic fields the solution of the equations of motion gives²⁵

$$\begin{aligned} x &= \frac{E}{\omega_c B} (1 - \cos \omega_c t), \\ y &= -\frac{E}{\omega_c B} (\omega_c t - \sin \omega_c t), \\ z &= 0 \end{aligned} \quad (12.25)$$

for the initial conditions $x = y = z = \dot{x} = \dot{y} = \dot{z} = 0$. The trajectory

²⁴As shown in Fig. 1.6 for the classical model of the Zeeman effect. This assumes zero velocity along the z -axis. See also Blundell (2001).

²⁵See Bleaney and Bleaney (1976, Problem 4.10).

of the ion is a cycloid: a combination of circular motion at ω_c and drift at velocity E/B in the y -direction,²⁶ as illustrated in Fig. 12.8(c). The counter-intuitive drift of the ion's average position in the direction *perpendicular* to $\mathbf{E}$ is the key to the operation of the Penning trap. The drift of the ion perpendicular to the radial electric field gives a tangential component of velocity and causes the ion to move slowly around the z -axis (direction of $\mathbf{B}$) at the magnetron frequency ω_m whilst at the same time undergoing cyclotron orbits,²⁷ as shown in Fig. 12.8(d). The electrode structure shown in Fig. 12.7 gives a radial field in the plane $z = 0$. In addition to ω_c and ω_m , the ion's motion has a third characteristic frequency ω_z associated with oscillations along the z -axis of the trap (analogous to the axial motion between the two d.c. electrodes at either end of the linear Paul trap). Usually the three frequencies have widely

²⁶It is easily seen that E/B has the dimensions of a velocity by looking at the expressions for the electric and magnetic forces given above.

²⁷In a magnetron, a beam of electrons in crossed E - and B -fields moves in a similar way to that shown in Fig. 12.8, but much faster than ions because of the smaller mass and higher electric field. These electrons radiate electromagnetic radiation at ω_m in the microwave region, e.g. at 2.5 GHz for the magnetrons in domestic microwave ovens (Bleaney and Bleaney 1976, Section 21.5).

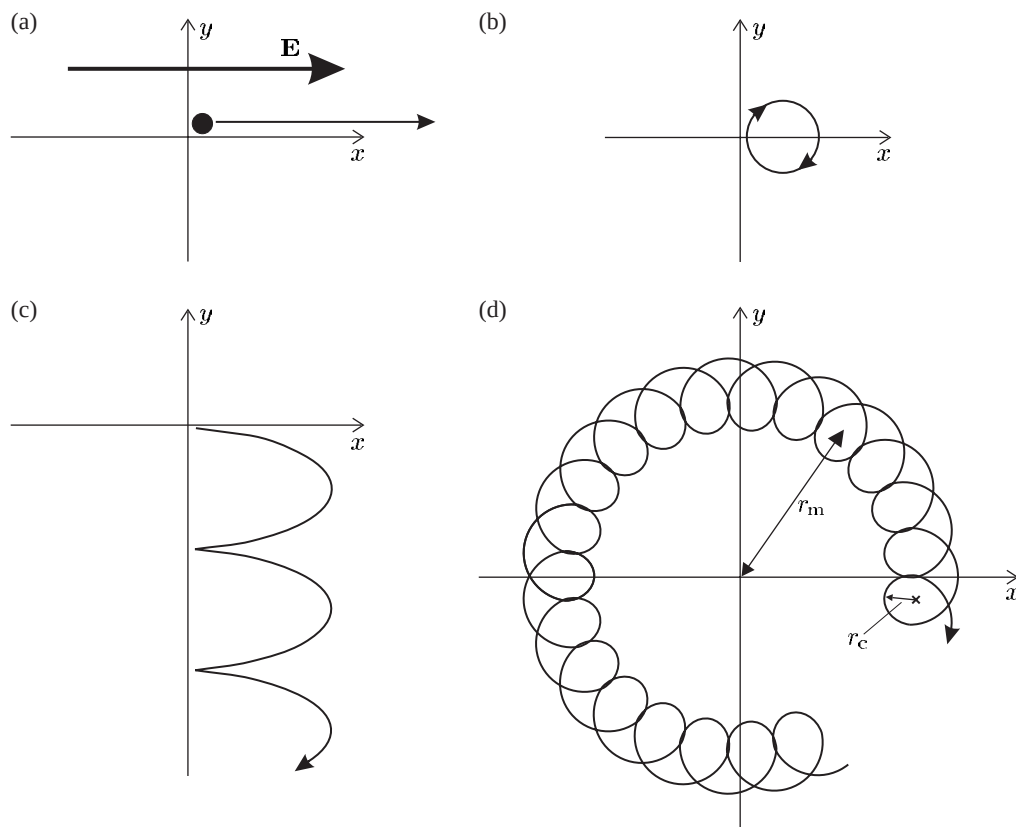


Fig. 12.8 The motion of a positively-charged ion in various configurations of electric and magnetic fields. (a) A uniform electric field along the x -direction accelerates the ion in that direction. (b) A uniform magnetic flux density $\mathbf{B}$ along the z -direction (out of the page) leads to a circular motion in the plane perpendicular to $\mathbf{B}$, at the cyclotron frequency ω_c . (c) In a region of crossed electric and magnetic fields (E_x and B_z , respectively) the motion described by eqns 12.25 is drift at velocity E/B perpendicular to the uniform magnetic field in addition to the cyclotron orbits. (The ion is initially stationary.) (d) In a Penning trap the combination of a radial electric field and axial magnetic field causes the ion to move around in a circle at the magnetron frequency. (If $B = 0$ then the ion would move radially outwards and hit the ring electrode.)

different values $\omega_z \ll \omega_m \ll \omega_c$ (Exercise 12.4).

12.7.2 Mass spectroscopy of ions

The determination of the mass of ions by means of eqn 12.20 for Paul trapping has been mentioned earlier. Alternatively, measurement of the ratio of cyclotron frequencies of two different species of ion in the same Penning trap gives their mass ratio:

$$\frac{\omega'_c}{\omega_c} = \frac{eB/M'}{eB/M} = \frac{M}{M'}. \quad (12.26)$$

This assumes the simplest case with two species of equal charge, but the ratio of the charges is always known exactly. Superconducting magnets give very stable fields so that the cancellation of B in the above equation introduces very little uncertainty and in this way masses can be compared to better than 1 part in 10^8 .

12.7.3 The anomalous magnetic moment of the electron

The advantages of the Penning trap have been exploited to make precise measurements of the magnetic moment of the electron (confined in the same way as ions but with a negative voltage on the end caps). From an atomic physics perspective, this can be viewed as a measurement of the Zeeman effect for an electron bound in a trap rather than one bound in an atom (Dehmelt 1990), but the splitting between the two magnetic states $m_s = \pm 1/2$ is the same in both situations, corresponding to a frequency $\Delta\omega = g_s \mu_B B / \hbar = g_s e B / 2m_e$. Measurement of this frequency gives the gyromagnetic ratio for spin g_s . To determine B accurately they measured the cyclotron frequency $\omega_c = eB/m_e$ and found the ratio

$$\frac{\Delta\omega}{\omega_c} = \frac{g_s}{2}. \quad (12.27)$$

The relativistic theory of quantum mechanics developed by Dirac predicts that g_s should be exactly equal to 2, but the incredibly precise measurement by Van Dyck *et al.* (1986) found

$$\frac{g_s}{2} = 1.0011596521884(4).$$

The accuracy is better than 4 in 10^{12} . Often this is quoted as a measurement of $g - 2$ for the electron and the difference from 2 arises from quantum electrodynamic (QED) effects. For the electron the theoretical calculation gives

$$\frac{g_s}{2} = 1 + \frac{\alpha}{2\pi} + A_2 \left(\frac{\alpha}{\pi}\right)^2 + A_3 \left(\frac{\alpha}{\pi}\right)^3 + A_4 \left(\frac{\alpha}{\pi}\right)^4 + \dots \quad (12.28)$$

The very detailed calculations give the coefficients $A_2 = -0.328478965$, $A_3 = 1.17611$ and $A_4 = -0.99$. The numerical value of this expression

agrees well with the experimental value given above and this provides a very stringent test of the theory.²⁸ Actually, at the time of the first measurements the fine-structure constant α was not known to enough decimal places, so the argument was turned around—the theory was assumed to be correct and used to deduce the value of α . Writing the theoretical value as coefficients multiplying various powers of α (in this case α/π) reflects the way theorists carry out QED calculations. Each power corresponds to perturbations of a given order. To match the accuracy of the experiment required evaluation of the contribution from all orders of perturbation up to and including the fourth order.²⁹ As might be imagined, it took many years of careful work to match the phenomenal precision of the experiment.

Similar experiments have also been carried out for the positron, the antimatter counterpart of the electron, and the comparison of the properties of particles and antiparticles gives interesting tests of the fundamental symmetry principles of particle physics. The very accurate theoretical calculations of the magnetic moments can only be made for simple particles without internal structure (leptons). Other QED experiments provide complementary information; for example, the measurements of the Lamb shift in hydrogen, and highly-ionized hydrogenic ions test the theory for an electron in a bound state where the calculations are considerably more complicated than for a free electron. It is very important to understand how to apply field theories like QED to bound systems.

12.8 Electron beam ion trap

The electron beam ion trap (EBIT) was developed to confine ions that have lost many electrons and which have energies much higher than those in typical experiments with Paul and Penning traps.

Figure 12.9 shows a schematic layout of an EBIT. Such a device is physically much larger than the other types of trap, but still much smaller than the particle accelerators that were previously used to produce highly-ionized ions.³⁰ The EBIT confines positive ions by their strong electrostatic attraction to the high negative charge density in an electron beam along the axis of the trap—the ions stay within this electron beam most of the time. The electrons emanate from an electron gun with a high current density over a small area, but the space charge in the beam tends to cause divergence. A strong axial magnetic field counteracts this spreading to keep the electrons tightly focused. This magnetic field acts as in a Penning trap to prevent the electrons moving radially outwards under the influence of the radial electric field—the same field that confines positive ions pushes electrons outwards.³¹ Electrodes with a d.c. voltage of several kilovolts restrict the motion of the ions along the electron beam. (In comparison, the Penning trap only has a few volts on the end caps.)

In addition to trapping, the EBIT also produces the ions by the following ionization steps.³² Atoms or ions in a low-charge state injected

²⁸In 1998 CODATA recommended the value of $\alpha = 1/137.0359979$. The latest value may be found on the NIST web site.

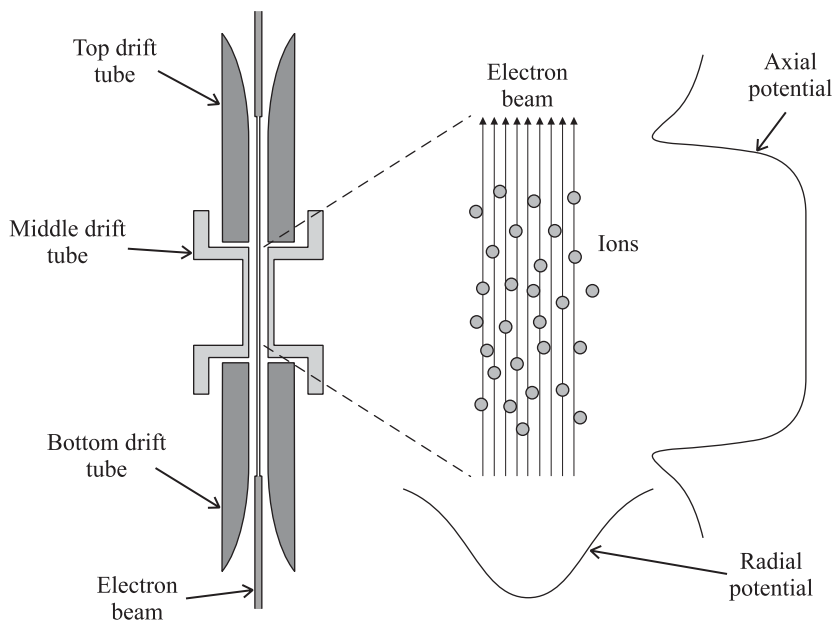
²⁹This required the evaluation of about 1000 contributions, each represented by different Feynman diagrams, and the calculations are still being refined by Kinoshita (1995) and Kinoshita and Nio (2003).

³⁰The EBIT uses a hybrid of the techniques in ion trapping and accelerator physics to extend the precision of traps to higher energies.

³¹The magnetic field has a negligible effect directly on the ions, in comparison to the electrostatic force from the space charge of the electrons.

³²It has the alternative name of an electron beam ion source or EBIS.

Fig. 12.9 A cross-section of an electron beam ion trap that has cylindrical symmetry. The high-energy electron beam along the axis of the trap (that is obviously a negative space charge) attracts positive ions to give radial confinement and ionizes them further. The electrodes (called drift tubes by accelerator physicists) give confinement along the axis, i.e. the top and bottom drift tubes act like end caps as in a Penning trap but with much higher positive voltage. To the right is an enlarged view of the ions in the electron beam and the form of the electrostatic potentials along the radial and axial directions. Courtesy of Professor Joshua Silver and co-workers, Physics department, University of Oxford.



into the EBIT region have electrons knocked off by the electron beam to form positive ions. These ions become confined within the electron beam where bombardment by the high-energy electrons removes more and more electrons, so that the ions become more highly charged. This process continues until the electrons remaining on the ion have a binding energy greater than the energy of the incoming electrons. Thus the final charge state of the trapped ion is controlled by varying the accelerating voltage on the electron gun. As an extreme example, consider stripping all but one of the electrons off a uranium atom. The final stage of the ionization process to produce U^{+91} requires an energy of $13.6 \times (92)^2 \sim 10^5$ eV, i.e. an accelerating voltage of 100 kV. These extreme conditions can be achieved but many EBIT experiments use more modest voltages on the electron gun of a few tens of kilovolts.

The transitions between the energy levels of highly-charged ions produce X-rays and the spectroscopic measurements of the wavelength of the radiation emitted from EBITs, use vacuum spectrographs (often with photographic film as the 'detector' since it has a high sensitivity at short wavelengths and gives good spatial resolution). Such traditional spectroscopic methods have lower precision than laser spectroscopy but QED effects scale up rapidly with increasing atomic number. The Lamb shift increases as $(Z\alpha)^4$ whereas the gross energy scales as $(Z\alpha)^2$, so measurements of hydrogen-like ions with high Z allow the QED effects to be seen. It is important to test the QED calculations for bound states because, as mentioned in the previous section, they require distinctly different approximations and theoretical techniques to those used for free particles. Indeed, as Z increases the parameter $Z\alpha$ in the expansions

used in the QED calculations becomes larger and higher orders start to make a greater contribution.

12.9 Resolved sideband cooling

Laser cooling on a strong transition of line width Γ rapidly reduces the energy of a trapped ion down to the Doppler cooling limit $\hbar\Gamma/2$ of that transition. The laser cooling of an ion works in a very similar way to the Doppler cooling of a free atom (see Exercise 9.8). To go further narrower transitions are used. However, when the energy resolution of the narrow transition $\hbar\Gamma'$ is less than the energy interval $\hbar\omega_v$ between the vibrational levels of the trapped ion (considered as a quantum harmonic oscillator) the quantisation of the motion must be considered, i.e. in the regime where

$$\Gamma' \ll \omega_v \ll \Gamma. \quad (12.29)$$

We have seen that typically $\Gamma/2\pi$ is tens of MHz and $\omega_v/2\pi \simeq 1$ MHz. The vibrational energy levels have the same spacing $\hbar\omega_v$ in both the ground and excited states of the ion since the vibrational frequency depends only on the charge-to-mass ratio of the ion and not its internal state, as illustrated in Fig. 12.10. The trapped ion absorbs light at the angular frequency of the narrow transition for a free ion ω_0 , and also at the frequencies $\omega_L = \omega_0 \pm \omega_v$, $\omega_0 \pm 2\omega_v$, $\omega_0 \pm 3\omega_v$, etc. that correspond to transitions in which the vibrational motion of the ion changes. The vibrational levels in the ground and excited states are labelled by the vibrational quantum numbers v and v' , respectively, and these sidebands correspond to transitions in which $v' \neq v$. The energy of the bound system is reduced by using laser light at frequency $\omega_L = \omega_0 - \omega_v$ to excite the first sideband of lower frequency, so that the ion goes into the vibrational level with $v' = v - 1$ in the upper electronic state. This excited state decays back to the ground state—the most probable spontaneous transition is the one in which the vibrational level does not change so that, on average, the ion returns to the ground electronic state in a lower vibrational level than it started. A detailed explanation of the change in v during spontaneous emission would need to consider the overlap of the wavefunctions for the different vibrational levels and is not given here.³³

This sideband cooling continues until the ion has been driven into the lowest vibrational energy level. An ion in the $v = 0$ level no longer absorbs radiation at $\omega_0 - \omega_v$, as indicated in Fig. 12.10—experiments use this to verify that the ion has reached this level by scanning the laser frequency to observe the sidebands on either side, as shown in Fig. 12.10(b); there is little signal at the frequency of the lower sideband, as predicted, but there is a strong signal at the frequency of the upper sideband.³⁴ The lower sideband arises for the population in the $v = 1$ level (and any higher levels if populated), and the upper sideband comes from ions in $v = 0$ that make a transition to the $v' = 1$ level. Thus the ratio of the signals on these two sidebands gives a direct measure of the

³³The situation closely resembles that in the Franck–Condon principle that determines the change in vibrational levels in transitions between electronic states of molecules—the relevant potentials for an ion, shown in Fig. 12.10, are simpler than those for molecules.

³⁴It is easy to be misled into thinking that the number of sidebands observed depends on the occupation of the vibrational levels of the trapped ion (especially if you are familiar with the vibrational structure of electronic transitions in molecules). This example shows that this is not the case for ions, i.e. sidebands arise for an ion that is predominantly in the lowest vibrational level. Conditions can also arise where there is very weak absorption at the frequencies of the sidebands, even though many vibrational levels are occupied, e.g. for transitions whose wavelength is much larger than the region in which the ions are confined. Although sidebands have been explained here in terms of vibrational levels, they are not a quantum phenomenon—there is an alternative classical explanation of sidebands in terms of a classical dipole that emits radiation while it is vibrating to and fro.

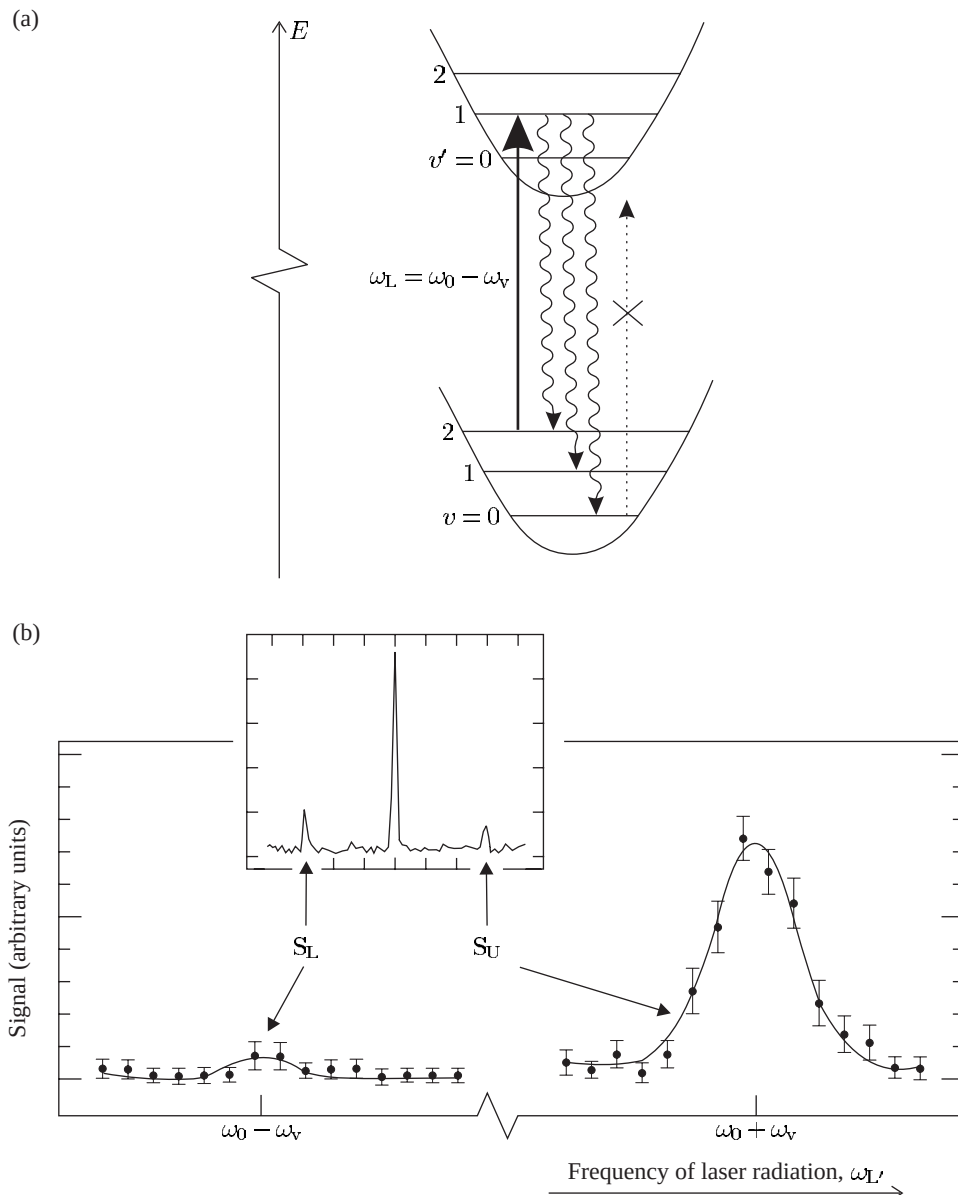


Fig. 12.10 (a) The vibration levels of an ion in a harmonic trap have the same spacing for both the ground and excited electronic states. Excitation by light of frequency $\omega_L = \omega_0 - \omega_v$, followed by spontaneous emission, leads to a decrease in the vibrational quantum number, until the ion reaches the lowest level with $v = 0$, from whence there is no transition, as indicated by the dotted line at this frequency (but light at frequency ω_0 and $\omega_0 + \omega_v$ would excite the ion in $v = 0$). (b) The spectrum of a trapped ion shows sidebands on either side of the main transition. The inset figure shows the spectrum before sideband cooling. The enlarged part of the figure shows the signal on the upper sideband S_U at angular frequency $\omega_0 + \omega_v$ is stronger than the lower one S_L at $\omega_0 - \omega_v$ —this asymmetry indicates that the ion is mainly in the lowest vibrational state (i.e. vibrational quantum number much less than 1). The vertical axis gives the transition probability, or the probability of a quantum jump during the excitation of the narrow transition. After Diedrich *et al.* (1989). Part (b), copyright 1989 by the American Physical Society.

ratio of populations: in this example $\mathcal{N}(v=1)/\mathcal{N}(v=0) = 0.05$ so the ion spends most of its time in the lowest level. Thus the ion has almost the minimum energy attainable in this system.

Single ion experiments such as optical frequency standards with extremely high resolution (Section 12.6) do not need to cool the ion to the very lowest level—they just pick out the transition at ω_0 from the well-resolved sidebands. The quantum computing experiments described in the next chapter, however, must have an initial state with all the ions in the lowest energy level of the trap (or very close to this ideal situation) to give complete control over the quantum state of the whole system. The preparation of all the trapped ions in the lowest vibrational level³⁵ is complicated by the collective modes of vibration of a system with more than one trapped ion (Exercise 12.1), and the achievement of this initial state stretches the capability of laser cooling methods to their very limits.³⁶

12.10 Summary of ion traps

This chapter explored some of the diverse physics of ion trapping, ranging from the cooling of ions to temperatures of only 10^{-3} K in small ion traps to the production of highly-charged ions in the EBIT. Trapping of positrons was mentioned in Section 12.7.3 and ion traps make good containers for storing other types of antimatter such as antiprotons produced at particle accelerators.³⁷ In recent experiments at CERN carried out by a large collaboration (Amoretti *et al.* 2002) these two antiparticles have been put together to produce anti-hydrogen. In the future it will be possible to do anti-atomic physics, e.g. to measure whether hydrogen and anti-hydrogen have the same spectra (a test of CPT invariance.) This high-energy trapping work has developed from accelerator-based experiments and probes similar physics.

At the opposite pole lies the work on the laser cooling of ions to extremely low energies. We have seen that the fundamental limit to the cooling of a bound system is quite different to the laser cooling of free atoms. Experimenters have developed powerful techniques to manipulate single ions and make frequency standards of extreme precision. The long decoherence times of trapped ions are now being exploited to carry out the manipulation of several trapped ions in experiments on quantum computation, which is the subject of the next chapter. Such experimental techniques give exquisite control over the state of the whole quantum system in a way that the founders of quantum mechanics could only dream about.

Further reading

The book on ion trapping by Ghosh (1995) gives a detailed account of these techniques. See also the tutorial articles by Wayne Itano (Itano *et al.* 1995) and David Wineland (Wineland *et al.* 1995), and the Nobel

³⁵For neutral particles in magnetic traps, quantum statistics causes the atoms to undergo Bose–Einstein condensation into the ground state, even though they have a mean thermal energy greater than the spacing between trap energy levels. Quantum statistics does not affect trapped ions because they are distinguishable—even if the ions are identical the mutual Coulomb repulsion keeps them far apart, as shown in Fig. 12.4, and the strong fluorescence enables the position of each ion to be determined.

³⁶The alert reader may have noticed that we have not discussed the recoil limit, that plays such an important role for free particles. For stiff traps the spacing of the vibrational energy levels greatly exceeds the recoil energy $\hbar\omega_v \gg E_{\text{rec}}$, and the cooling limit of the trapped particle is determined by the zero-point energy.

³⁷Just after its creation in high-energy collisions, the antimatter has an energy of MeV but it is moderated to energies of keV before trapping.

- ³⁸On the web site of the Nobel prizes. prize lecture of Wolfgang Paul.³⁸ The National Physical Laboratory in the UK and the National Institute of Standards and Technology in the US provide internet resources on the latest developments and research.

Exercises

- (12.1) *The vibrational modes of trapped ions*
Two calcium ions in a linear Paul trap lie in a line along the z -axis.

- (a) The two end-cap electrodes along the z -axis produce a d.c. potential as in eqn 12.23, with $a_2 = 10^6 \text{ V m}^{-2}$. Calculate ω_z .
- (b) The displacements z_1 and z_2 of the two ions from the trap centre obey the equations

$$M\ddot{z}_1 = -M\omega_z^2 z_1 - \frac{e^2/4\pi\epsilon_0}{(z_2 - z_1)^2},$$

$$M\ddot{z}_2 = -M\omega_z^2 z_2 + \frac{e^2/4\pi\epsilon_0}{(z_2 - z_1)^2}.$$

Justify the form of these equations and show that the centre of mass, $z_{\text{cm}} = (z_1 + z_2)/2$ oscillates at ω_z .

- (c) Calculate the equilibrium separation a of two singly-charged ions.
- (d) Find the frequency of small oscillations of the relative position $z = z_2 - z_1 - a$.
- (e) Describe qualitatively the vibrational modes of three ions in the trap, and the relative order of their three eigenfrequencies.³⁹

- (12.2) *Paul trap*

- (a) For Hg^+ ions in a linear Paul trap with dimensions $r_0 = 3 \text{ mm}$, calculate the maximum amplitude V_{max} of the radio-frequency voltage at $\Omega = 2\pi \times 10 \text{ MHz}$.
- (b) For a trap operating at a voltage $V_0 = V_{\text{max}}/\sqrt{2}$, calculate the oscillation frequency of an Hg^+ ion. What happens to a Ca^+ ion when the electrodes have the same a.c. voltage?
- (c) Estimate the depth of a Paul trap that has $V_0 = V_{\text{max}}/\sqrt{2}$, expressing your answer as a fraction of eV_0 .

- (d) Explain why a Paul trap works for both positive and negative ions.

- (12.3) *Investigation of the Mathieu equation*

Numerically solve the Mathieu equation and plot the solutions for some values of q_x . Give examples of stable and unstable solutions. By trial and error, find the maximum value of q_x that gives a stable solution, to a precision of two significant figures. Explain the difference between precision and accuracy. [*Hint.* Use a computer package for solving differential equations. The method in Exercise 4.10 does not work well when the solution has many oscillations because its numerical integration algorithm is too simple.]

- (12.4) *The frequencies in a Penning trap*

A Penning trap confines ions along the axis by repulsion from the two end-cap electrodes; these have a d.c. positive voltage for positive ions that gives an axial oscillation frequency, as calculated in Exercise 12.1. This exercise looks at the radial motion in the $z = 0$ plane. The electrostatic potential in eqn 12.23 with $a_2 = 10^5 \text{ V m}^{-2}$ leads to an electric field that points radially outwards, but the ion does not fly off in this direction because of a magnetic field of induction $B = 1 \text{ T}$ along the z -axis.

Consider a Ca^+ ion.

- (a) Calculate the cyclotron frequency.
- (b) Find the magnetron frequency. [*Hint.* Work out the period of an orbit of radius r in a plane perpendicular to the z -axis by assuming a mean tangential velocity $v = E(r)/B$, where $E(r)$ is the radial component of the electric field at r .]

³⁹They resemble the vibrations of a linear molecule such as CO_2 , described in Appendix A; however, a quantitative treatment would have to take account of the Coulomb repulsion between all pairs of ions (not just nearest neighbours).

(12.5) *Production of highly-charged ions in an EBIT*

- (a) Estimate the accelerating voltage required for an electron beam voltage that produces hydrogenic silicon Si^{13+} in an EBIT.
- (b) Calculate the radius of the first Bohr orbit ($n = 1$) in hydrogenic uranium, U^{+91} .
- (c) Calculate the binding energy of the electron in U^{+91} and express it as a fraction of the atom's rest mass energy Mc^2 .
- (d) QED effects contribute $3 \times 10^{-5} \text{ eV}$ to the

binding energy of the 1s ground configuration of atomic hydrogen. Express this as a fraction of the hydrogen atom's rest mass. Estimate the magnitude of the QED contribution in the ground state of hydrogen-like uranium U^{+91} as a fraction of the rest mass. This fraction gives the precision $\Delta M/M$ with which the ion's mass must be determined in order to measure QED effects. Discuss the feasibility of doing this in an ion trap.

Web site:

<http://www.physics.ox.ac.uk/users/foot>

This site has answers to some of the exercises, corrections and other supplementary information.